

Peer Response 1

[Fahad](#)

Fahad, your post provides a straightforward and elaborate description of how Agent Communication Languages (ACLs) like KQML can achieve meaningful communication within multi-agent systems. I concur with you that the key strength of ACLs is that they express semantics, not syntax, and agents can share not only information but also their plans, intentions, and commitments. The difference is the key to cooperation within dynamic and open settings. The significance of structured, semantically rich communication in enhancing collaborative action and level of reasoning is also included in current studies on multimodal large language model-oriented agents (Xie et al., 2024).

I was particularly interested in how you emphasised the aspect of heterogeneity. As you clarified, ACLS enables agents that may be programmed in different languages or platforms to interoperate using a single standard. This capability is essential to distributed applications and systems such as robotics and autonomous driving, where agents must cooperate irrespective of their design differences. The recent research on multimodal large language models in autonomous driving emphasises the importance of interoperability in integrating perception, decision-making, and interaction among multiple systems (Cui et al., 2024). ACLs are capable of that common ground, so they help scale complicated systems.

Simultaneously, I would like to concur with your discussion of constraints. Shared ontologies and sophisticated reasoning add to computational overhead, making ACLs less effective in real-time applications. This is consistent with recent results on conversational agents, which note that semantic richness is both a benefit and a performance issue (Ma et al., 2024). This trade-off between invoking a lightweight method to be fast and ACLs to be flexible brings you to the essence of the design dilemma in constructing contemporary multi-agent systems.

All in all, your post offers a rather balanced view, demonstrating the feasibility problems and longevity of the ACLs in research and system creation.

References

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Peer Response 2

[Ali Alhammadi](#)

Ali, in your post, it is evident how the Agent Communication Languages (ACLs) and specifically KQML facilitate agent-agent communication by emphasising knowledge and intention transfer, not mere message transmission. I agree that the difference makes multi-agent systems (MAS) effective in open and heterogeneous environments, where flexibility and interoperability are needed. Recent surveys on multi-agent systems based on LLMs confirm that structured communication is among the most significant aspects of letting the agents cooperate and adapt to the changing environment (Li et al., 2024).

It is also significant how you highlight standardization as a strength of ACLs. As you described, distributed applications like e-commerce, supply chain management, and knowledge-sharing must be compatible with various systems. ACLs improve scalability and interoperability because they deliver a common communication layer. This aligns with the larger research patterns that emphasise that agents engineered by other organizations can work together more effectively if communication protocols become standardised (Liu et al., 2023).

Meanwhile, your trade-offs were correctly recognized. ACLs need high reasoning, which makes them complex and slow to compute. They are also unsuitable in real-time or performance-sensitive systems, such as robotics, because parsing and interpreting performatives have overhead. These disadvantages make lightweight methods like Python or Java method invocation a favorite in industries because they are fast and efficient. However, as you observed, these methods are too finite to provide the abstraction and independence required in loosely coupled systems where agents must decode one another.

Lastly, it is clear why ACLs continue to dominate the research world, which is why they are so significant. Simpler variants prevail in the commercial arena; however, ACLs still directly impact the conceptualisation of interoperability and intelligent system design in academia and professional environments (Getchell et al., 2022).

References

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Peer Response 3

Abdulla Almessabi

Abdulla, your post gives an excellent introduction to the role of Agent Communication Languages (ACLs), including KQML as a high-level communication protocol in multi-agent systems. I also concur with your argument that their lack of dependence on implementation languages is particularly what makes ACLs useful in heterogeneous and distributed environments, where the agents must be able to coordinate with each other despite being implemented on different platforms. This aligns with recent studies demonstrating how structured communication is vital with larger language model-based systems taking a central role in intelligent applications (Fan et al., 2024).

I also enjoy how you touch upon the fact that ACLs can accommodate more interactive tasks, such as negotiation, cooperation, and delegation of tasks. These upper-level performatives are more important to achieve intelligent agents that are more than data exchange machines since they facilitate joint work and mutual decision-making. This view is further supported by modern research on large language models, which states that communication acts are the source of higher-order reasoning and coordination in multi-agent systems (Su et al., 2024).

Simultaneously, you also made the weaknesses of ACLs quite visible. They are expressive but inefficient; without consensus on ontologies, it is hard to have real semantic interoperability. This corresponds to the results that language models and communication protocols provide flexibility and abstraction, but, at the same time, may also raise the cost of computation and generate ambiguity unless designed well (Hao et al., 2022). It was also appropriate when you compared it to method invocation in Python or Java, demonstrating that, although method calls offer a direct and efficient interaction in tightly-coupled systems, they cannot possibly be as open and interoperable as ACLs are in a distributed system.

Overall, your post provides a good balance of the strong and the weak in ACLs. You demonstrate that the choice of using ACLs or method invocation is mainly determined by the system's emphasis on heterogeneity, flexibility, efficiency, and simplicity.

References:

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